SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

*I **_S.HARIPRIYA** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

Annexure

Question 1: File Transfer — Segments, Frames, Overhead & Time

Given

- File size = 150 MB
- Bandwidth = 50 Mbps
- Transport Layer segment size = 1500 bytes
- Data Link Layer adds 40 bytes (header + trailer) per frame

Step 1: Convert File Size to Bytes

150 MB = 150 × 1,000,000 = 150,000,000 bytes

(Using 1 MB = 10^6 bytes, the standard convention for networking bandwidth problems.)

Step 2: Number of Segments (Transport Layer)

Segments = 150,000,000 ÷ 1500 = 100,000 segments

Each segment becomes exactly one frame at the Data Link layer, so Frames = 100,000.

Step 3: Data Link Layer Overhead

Each frame adds 40 bytes of header + trailer:

Total Overhead = 100,000 × 40 = 4,000,000 bytes = 4 MB

Step 4: Total Data Actually Transmitted

Total Data = 150 MB (payload) + 4 MB (overhead) = 154 MB

Converted to bits: 154,000,000 bytes × 8 = 1,232,000,000 bits

Step 5: Transmission Time (Ignoring Propagation Delay)

Time = 1,232,000,000 bits ÷ 50,000,000 bps = 24.64 seconds

Summary Table

Parameter	**Value**
Total Segments	100,000
Total Frames	100,000
Data Link Layer Overhead	4,000,000 bytes (4 MB)
Total Transmitted Data	154 MB
Transmission Time	24.64 seconds

Explanation: How Transport and Data Link Layers Ensure Reliable Communication

- Transport Layer — Segmentation: breaks the 150 MB file into 1500-byte segments so lower layers can handle them individually.
- Transport Layer — Sequencing: numbers each segment so the receiver reassembles the file in correct order even if segments arrive out of sequence.
- Transport Layer — Acknowledgment & Retransmission: receiver sends ACKs; unacknowledged segments are retransmitted, which is what makes protocols like TCP reliable.
- Transport Layer — Flow control: paces transmission so the sender doesn't overwhelm the receiver's buffer.
- Data Link Layer — Framing: wraps each segment with a header (MAC addressing) and trailer (CRC) for the physical hop.
- Data Link Layer — Error detection: CRC in the trailer lets the receiver detect corrupted frames and discard/request them again.
- Data Link Layer — Node-to-node scope: guarantees reliable delivery only across a single physical link, unlike Transport's end-to-end guarantee.

Together, the Transport layer guarantees the file arrives complete and in order end-to-end, while the Data Link layer guarantees each individual hop is free of transmission errors.

Question 2 & 4: Sliding Window — Transmission Windows and Retransmissions (duplicate question)

Given

- Sliding window size = 6 frames
- Each frame carries 1024 bytes of user data
- Sender transmits 48 frames
- One frame lost per window

Step 1: Number of Transmission Windows

Windows = Total Frames ÷ Window Size = 48 ÷ 6 = 8 windows

Step 2: Total Retransmissions

1 lost frame per window × 8 windows = 8 retransmissions

Step 3: Total Data Successfully Delivered (for context)

48 frames × 1024 bytes = 49,152 bytes of user data across all windows

Summary Table

Parameter	Value
Frames per Window	6
Total Frames Sent	48
Total Windows	8
Frames Lost	8 (1 per window)
Total Retransmissions	8

Explanation: How Flow Control Improves Communication Efficiency

- Sliding window flow control allows the sender to transmit multiple frames (a full window) before waiting for an acknowledgment, instead of stopping after every single frame as in stop-and-wait protocols.
- This keeps the communication link continuously busy, significantly increasing throughput compared to sending one frame and waiting for its ACK each time.
- The receiver still controls the window size, so it can throttle the sender if its buffer is filling up — preventing data loss due to buffer overflow.

- When a frame within a window is lost, only the affected frame(s) need retransmission (in protocols like Selective Repeat), minimizing wasted bandwidth compared to resending the entire window.
- Overall, flow control balances speed (larger windows = more throughput) against reliability (receiver capacity and loss handling).

Question 3: IP Fragmentation — Fragments and Header Overhead

Given

- IP packet size = 12,000 bytes
- Maximum Transmission Unit (MTU) = 1500 bytes
- IP header per fragment = 20 bytes

Step 1: Usable Payload per Fragment

Payload per fragment = MTU − Header = 1500 − 20 = 1480 bytes

Step 2: Number of Fragments

Fragments = $\lceil 12{,}000 \div 1480 \rceil = \lceil 8.108 \rceil = 9$ fragments

(8 fragments carry 1480 bytes each = 11,840 bytes; the 9th carries the remaining 160 bytes.)

Step 3: Total Header Overhead

Overhead = 9 fragments × 20 bytes = 180 bytes

Summary Table

Parameter	Value
Payload per Fragment	1480 bytes
Total Fragments	9
Total Header Overhead	180 bytes
Total Data Transmitted	12,000 + 180 = 12,180 bytes

Explanation: How the Network Layer Ensures Successful Delivery of Fragmented Packets

- The Network layer fragments an oversized packet whenever it must pass through a link whose MTU is smaller than the packet size.
- Each fragment carries its own IP header containing the fragment offset (position of this fragment's data within the original packet) and flags (More Fragments / Don't Fragment) so the destination knows how to reassemble them.
- Fragmentation and reassembly happen once at the source/intermediate router and once at the final destination host — intermediate routers do not reassemble fragments, only the destination does.

- If any single fragment is lost, the entire original packet must typically be retransmitted (in IPv4), which is why minimizing unnecessary fragmentation is important for efficiency.
- This mechanism allows large packets to still travel across heterogeneous networks with differing MTU sizes, without the sender needing prior knowledge of every link's MTU along the path.

Question 5: Session Layer Synchronization Checkpoints

Given

- Video file size = 600 MB
- Checkpoint inserted every 60 MB of transmitted data
- Failure occurs after 390 MB transferred

Step 1: Number of Checkpoints Created Before Failure

Checkpoints = $\lfloor 390 \div 60 \rfloor = 6$ checkpoints

(Checkpoints occur at 60, 120, 180, 240, 300, and 360 MB.)

Step 2: Data to Be Retransmitted After Recovery

Data since last checkpoint = $390 - 360 = 30$ MB

Summary Table

Parameter	Value
Checkpoint Interval	60 MB
Data Transferred Before Failure	390 MB
Checkpoints Reached	6 (at 360 MB)
Data to Retransmit	30 MB

Explanation: The Importance of Synchronization in the Session Layer

- The Session layer establishes, manages, and terminates communication sessions between two systems, and one of its key responsibilities is synchronization via checkpoints.
- By inserting checkpoints periodically, the protocol allows a failed transfer to resume from the last confirmed checkpoint instead of restarting the entire 600 MB transfer from zero.
- In this scenario, only 30 MB (the data transferred since the last checkpoint at 360 MB) needs to be retransmitted after recovery — a huge saving compared to retransmitting all 390 MB or the full file.
- This is especially valuable for large multimedia transfers, video conferencing, and file backups, where restarting from scratch after any interruption would be highly inefficient and time-consuming.

- Synchronization also helps maintain session state consistency between communicating systems, which is essential for orderly recovery after crashes or network failures.

Question 6: Presentation Layer — Compression and Encryption Overhead

Given

- Original file size = 900 MB
- Presentation Layer compression = 40% reduction
- Encryption adds 5% overhead to the compressed size

Step 1: Compressed File Size

Compressed Size = $900 \times (1 - 0.40) = 900 \times 0.60 = 540$ MB

Step 2: Encrypted File Size

Encrypted Size = $540 \times (1 + 0.05) = 540 \times 1.05 = 567$ MB

Step 3: Total Percentage Reduction vs. Original

Reduction % = $(900 - 567) \div 900 \times 100 = 333 \div 900 \times 100 \approx 37.0\%$

Summary Table

Parameter	Value
Original Size	900 MB
Compressed Size (after 40% reduction)	540 MB
Encrypted Size (after +5% overhead)	567 MB
Net Reduction vs. Original	≈ 37.0%

Explanation: How the Presentation Layer Improves Transmission Efficiency and Security

- The Presentation layer is responsible for data representation, compression, and encryption — translating application data into a format suitable for transmission.
- Compression reduces the volume of data that must travel across the network, directly reducing transmission time and bandwidth consumption — here, cutting the file nearly in half (540 MB from 900 MB).
- Encryption then secures this compressed data against eavesdropping or tampering, though it necessarily adds some overhead (here, 5%) since encrypted data typically includes padding, initialization vectors, or authentication tags.

- Even with the added encryption overhead, the net effect is still a substantial 37% reduction in transmitted size compared to sending the original, uncompressed, unencrypted file.
- This illustrates the layer's dual role: making transmission more efficient (via compression) while also making it more secure (via encryption), with a small efficiency trade-off for that security.

Question 7: FTP Application — Requests and Transmission Time

Given

- Total data to transfer = 3 GB
- Effective throughput = 120 Mbps
- Application Layer divides data into 3 MB requests

Step 1: Convert Data to MB

3 GB = 3000 MB

Step 2: Total Number of Requests

Requests = 3000 MB ÷ 3 MB = 1000 requests

Step 3: Total Transmission Time

Total bits = 3000 MB × 8 = 24,000 Mb

Time = 24,000 Mb ÷ 120 Mbps = 200 seconds

Summary Table

Parameter	Value
Total Data	3 GB (3000 MB)
Request Size	3 MB
Total Requests	1000
Transmission Time	200 seconds

Explanation: How the Application Layer Provides Reliable Services to Users

- The Application layer (through protocols like FTP) provides the user-facing interface for file transfer, breaking a large 3 GB transfer into 1000 smaller, manageable requests.
- By dividing data into discrete requests, FTP allows better tracking of transfer progress and the ability to resume or retry individual failed requests instead of restarting the entire 3 GB transfer.
- FTP relies on the underlying Transport layer (TCP) to guarantee that each request's data arrives completely, in order, and without corruption — the Application layer itself doesn't handle low-level retransmission.

- This layered design lets users experience simple, reliable file transfer (upload/download commands) while the complexity of segmentation, acknowledgment, and error recovery is handled transparently by lower layers.
- Estimating transmission time (200 seconds here) also helps users and administrators plan for large transfers and assess whether current network throughput is adequate for their needs.

Question 8: End-to-End Delay Across OSI Layers

Given

- Application Layer processing delay = 4 ms
- Transport Layer processing delay = 3 ms
- Network Layer processing delay = 5 ms
- Data Link Layer processing delay = 2 ms
- Physical Layer processing delay = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Step 1: Total Processing Delay Across All Layers

Processing Delay = 4 + 3 + 5 + 2 + 1 = 15 ms

Step 2: Total End-to-End Delay

Total Delay = Processing Delay + Propagation Delay + Transmission Delay

Total Delay = 15 + 18 + 12 = 45 ms

Summary Table

Parameter	Value
Total Processing Delay (all 5 layers)	15 ms
Propagation Delay	18 ms
Transmission Delay	12 ms
Total End-to-End Delay	45 ms

Explanation: How Each Layer Contributes to the Communication Process

- Application Layer (4 ms): time spent preparing and formatting the user's data/request before handing it to the Transport layer.
- Transport Layer (3 ms): time spent segmenting data, adding sequence numbers, and preparing reliability mechanisms (checksums, port numbers).
- Network Layer (5 ms): time spent on logical addressing (IP) and routing decisions — typically the highest processing delay since routing/lookup can be computationally heavier.
- Data Link Layer (2 ms): time spent framing the packet with MAC addressing and computing error-checking codes (CRC).

- Physical Layer (1 ms): minimal processing time to convert frames into the actual bit stream / signal for the medium.
- Propagation delay (18 ms): time for the signal to physically travel across the medium, dependent on distance and signal speed — independent of processing at any layer.
- Transmission delay (12 ms): time to push all the packet's bits onto the link, dependent on packet size and link bandwidth.
- The total end-to-end delay (45 ms) is the sum of all these independent delay sources, showing how each OSI layer — plus the physical transmission and propagation characteristics of the link — together determine overall network latency.

Question 9: Frame Overhead and Transmission Efficiency

Given

- Useful data = 80 MB
- Frame size = 1600 bytes (includes overhead)
- Overhead per frame = 80 bytes

Step 1: Usable Payload per Frame

Payload per frame = 1600 − 80 = 1520 bytes

Step 2: Convert Data to Bytes

80 MB = 80,000,000 bytes

Step 3: Number of Frames Required

Frames = 80,000,000 ÷ 1520 = 52,631.58 → 52,632 frames (rounded up)

We round up because a partial frame still requires a full frame to be sent — you cannot send "0.58 of a frame."

Step 4: Total Overhead Transmitted

Total Overhead = 52,632 × 80 = 4,210,560 bytes ≈ 4.21 MB

Step 5: Transmission Efficiency

Efficiency = (Useful Data ÷ Total Data Transmitted) × 100

Total data transmitted = 52,632 × 1600 = 84,211,200 bytes

Efficiency = 80,000,000 ÷ 84,211,200 × 100 ≈ 95.0%

Summary Table

Parameter	Value
Payload per Frame	1520 bytes
Total Frames Required	52,632
Total Overhead	≈ 4.21 MB
Total Data Transmitted	≈ 84.21 MB
Transmission Efficiency	≈ 95%

Explanation: How Protocol Overhead Affects Network Performance

- Every frame carries “extra” bits beyond the actual user data — such as addressing, protocol type fields, sequence numbers, and error-checking codes — which is unavoidable since the network needs this metadata to correctly route, sequence, and verify each frame.
- More overhead generally means better reliability and easier troubleshooting, but it reduces the percentage of bandwidth carrying actual useful data — here, about 5% of all transmitted bits are overhead rather than payload.
- If frame size were smaller with the same fixed 80-byte overhead, efficiency would drop sharply, since overhead becomes a larger proportion of each frame; networks often use larger frames (Jumbo Frames) to push efficiency closer to 100%.
- This explains why protocols like Ethernet, Wi-Fi, and TCP/IP are designed with careful attention to header sizes — small inefficiencies multiply across millions of frames in high-throughput networks.
- A transmission efficiency of ~95% here is considered good, showing a reasonable balance between reliability overhead and useful throughput.

Question 10: Healthcare IoT — Multi-Layer Data Transfer

Given

- Medical record size = 240 MB
- Presentation Layer compression = 30%
- Transport Layer segment size = 1400 bytes
- Data Link Layer adds 60 bytes overhead per frame

Step 1: Compressed File Size (Presentation Layer)

Compressed Size = $240 \times (1 - 0.30) = 240 \times 0.70 = 168$ MB

Step 2: Number of Segments (Transport Layer)

168 MB = 168,000,000 bytes

Segments = $168{,}000{,}000 \div 1400 = 120{,}000$ segments (exact)

Step 3: Total Protocol Overhead (Data Link Layer)

Overhead = $120{,}000 \times 60 = 7{,}200{,}000$ bytes = 7.2 MB

Step 4: Final Data Transmitted Across the Physical Layer

Final Data = Compressed Size + DLL Overhead = 168 MB + 7.2 MB = 175.2 MB

Summary Table

Parameter	Value
Original Record Size	240 MB
Compressed Size (Presentation)	168 MB
Total Segments (Transport)	120,000
Total DLL Overhead	7.2 MB
Final Data over Physical Layer	175.2 MB

Explanation: How Multiple OSI Layers Work Together to Ensure Secure and Reliable Communication

- The Presentation layer first compresses the sensitive 240 MB medical record down to 168 MB, reducing the data volume that must be secured and transmitted — and in real healthcare systems, this layer would also typically encrypt the data for HIPAA-style compliance.

- The Transport layer then divides this compressed data into 120,000 segments of 1400 bytes each, enabling reliable, ordered delivery with acknowledgment and retransmission if any segment is lost — critical for medical data integrity.
- The Data Link layer frames each segment, adding 60 bytes of overhead per frame for addressing and error-checking (CRC), ensuring each hop across the IoT network is verified for correctness.
- The Physical layer ultimately transmits the final 175.2 MB (data + all added overhead) as raw bits across the healthcare network's physical medium (wired or wireless IoT links).
- This layered cooperation — compress and secure (Presentation), segment and guarantee delivery (Transport), frame and verify (Data Link), transmit (Physical) — ensures that critical patient data reaches its destination completely, correctly, and securely despite the resource constraints typical of IoT healthcare networks.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

		points			
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand